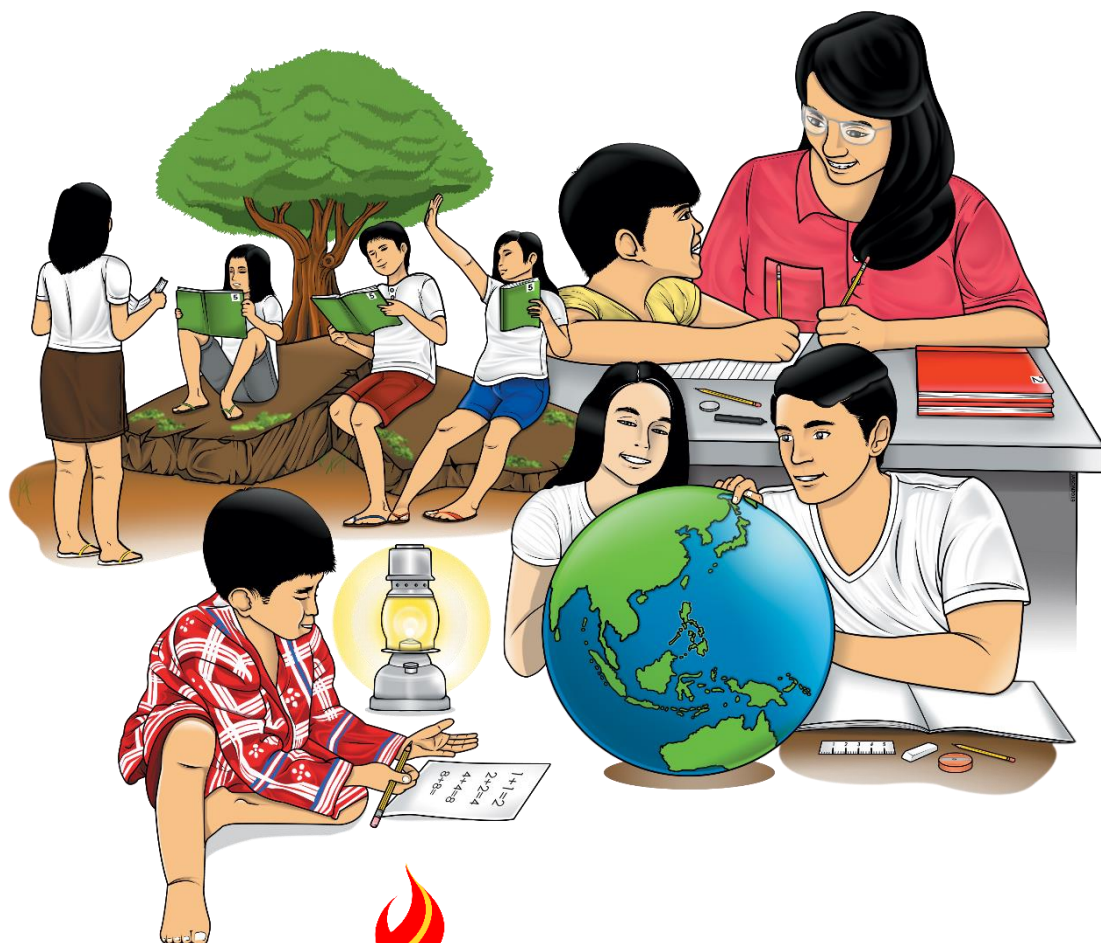


Mathematics

Quarter 2 – Module 2: Defining and Illustrating Ratio and Proportion



Mathematics– Grade 6
Alternative Delivery Mode
Quarter 2 – Module 2: Defining and Illustrating Ratio and Proportion
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Mathematics

Quarter 2 – Module 2: Defining and Illustrating Ratio and Proportion



What I Need to Know

This module was designed and written with you in mind. It is here to help you understand the concept of ratio and proportion. The scope of this module allows you to use it in many different learning situations. The language used recognizes your diverse vocabulary level.

The module contains lesson on defining and illustrating the meaning of ratio and proportion using concrete or pictorial models. **(M6NS-IIb-131)**

After going through this module, you are expected to:

1. define ratio and proportion;
2. use concrete objects or draw pictorial models to illustrate ratios and proportions; and
3. recognize when two quantities are in proportion.



What I Know

A. The table below shows the amounts of salt and sugar in proportion.

Amount of salt	3		9	12	
Amount of sugar	2	4			10

- 1) Complete the table.
- 2) Draw a block model that shows the amount of salt for 10 units of sugar.
- 3) Use the table to write three different proportions.

B. Read each statement carefully. Write TRUE or FALSE.

- _____ 1) $5 : 8 = 10 : 24$ is an example of a proportion.
- _____ 2) A proportion tells us that two ratios are equivalent.
- _____ 3) In a proportion, the quantities can be written in any order.
- _____ 4) The ratios $18 : 6$ and $3 : 1$ form a proportion.
- _____ 5) When the ratios of two quantities are different, the two quantities are not in the same proportion.

Lesson 1

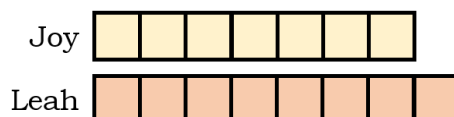
Defining and Illustrating Ratio and Proportion

A lot of what we do in real life involves making comparisons. Previously, you have learned how ratios and fractions are related and how they are used to compare two or more quantities. This time, you will be moving forward to understanding more deeply about ratio and proportion using concrete or pictorial models.



What's In

The model below compares the number of seashells Joy and Leah collected one day at the beach.



Write at least 5 statements about the model using fractions and ratios.

Example: The ratio of the number of seashells Joy collected to the number of seashells Leah collected is 7 .

1. _____
2. _____
3. _____
4. _____
5. _____



What's New

One hot day, Mother uses 2 ripe mangoes to make 3 glasses of mango shake. She plans to make 9 glasses. How many mangoes does she need?



What is It

Let us illustrate the problem.



For every 2 ripe mangoes used, 3 glasses of mango shake are made.



We can also say that the ratio of the number of mangoes used to the number of glasses of mango shake made is 2 : 3.

Find the number of ripe mangoes needed to make 9 glasses of mango shake.



The ratio of the number of mangoes needed to the number of glasses of mango shake is 6 : 9.

This means that Mother needs 6 ripe mangoes to make 9 glasses of mango shake.

The following table shows Mother's guide in preparing batches of mango shake.

Number of mangoes	2	4	6	8
Number of glasses	3	6	9	12
Mangoes : Glasses	2 : 3	4 : 6	6 : 9	8 : 12
Mangoes : Glasses (simplest form)	2 : 3	2 : 3	2 : 3	2 : 3

What can you say about the ratios?

The ratios are the same.

$$\frac{\text{number of mangoes}}{\text{number of glasses}} = \frac{2}{3} = \frac{4}{6}$$

$\xrightarrow{\times 2}$
 $\xleftarrow{\times 2}$

number of mangoes : number of glasses

$$\begin{array}{ccc} 2 & : & 3 \\ \times 2 \swarrow & & \searrow \times 2 \\ 4 & : & 6 \end{array}$$

When the first quantity doubles, the second quantity doubles also.

2 : 3 and 4 : 6 are equivalent ratios. These two ratios form a **proportion**.

2 : 3 = 4 : 6 can be read “2 is to 3 as 4 is to 6.”

$$\frac{\text{number of mangoes}}{\text{number of glasses}} = \frac{2}{3} = \frac{6}{9}$$

number of mangoes : number of glasses

$$2 : 3 = 6 : 9$$

When the first quantity triples, the second quantity triples also.

The ratios 2 : 3 and 6 : 9 are equivalent.

We can also say that 2 : 3 and 6 : 9 are in proportion.

From the table, what do you notice about the ratios 4 : 6 and 8 : 12?
Do they form a proportion? Why or why not?

Let's take a look at more examples.

Angelo prepares four different mixtures of orange paint. He then writes the ratio of the number of cans of red paint to the number of cans of yellow paint.

Mixture	A	B	C	D
Number of cans of red paint	3	6	8	12
Number of cans of yellow paint	1	3	5	9
$\frac{\text{Number of cans of red paint}}{\text{Number of cans of yellow paint}}$	$\frac{3}{1}$	$\frac{2}{1}$	$\frac{8}{5}$	$\frac{4}{3}$

What can you say about the fractions?

They are different.

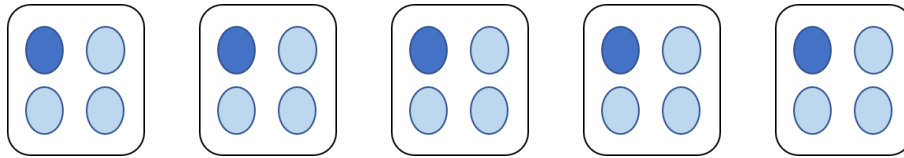
So, the numbers of cans of red paint and yellow paint are **not** in **proportion**.

There is 1 girl for every 3 boys in a dance group. There are 20 members altogether. How many girls and how many boys are there in the group?

Number of girls : Number of boys = 1 : 3

The ratio of the number of girls to the number of boys in the dance group is 1 : 3.

Method 1: Draw a picture



Method 2: Make a list

Number of girls	1	2	3	4	5
Number of boys	3	6	9	12	15
Total	4	8	12	16	20

Check:

$$\frac{\text{number of girls}}{\text{number of boys}} = \frac{1}{3} = \frac{5}{15}$$

$\xrightarrow{\times 5}$
 $\xleftarrow{\times 5}$

number of girls : number of boys

$$\begin{array}{ccc} 1 & : & 3 \\ \times 5 \swarrow & & \searrow \times 5 \\ 5 & : & 15 \end{array}$$

1 : 3 and 5 : 15 form a proportion.

Answer: There are 5 girls and 15 boys in the dance group.

Mrs Galvez uses 12 cups of water and 8 cups of oats to prepare oatmeal for her family. She mixes 6 cups of water and 4 cups of oats as extra.
Does Mrs Galvez use the same proportion for both batches of oatmeal?

Cups of water	12	6
Cups of oats	8	4
Water : Oats	3 : 2	3 : 2

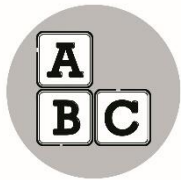
$$\frac{\text{number of cups of water}}{\text{number of cups of oats}} = \frac{12}{8} = \frac{6}{4}$$

$\xrightarrow{\div 2}$
 $\xleftarrow{\div 2}$

The ratios are the same.

The numbers of cups of water and oats are in proportion.

Answer: Yes. Mrs. Galvez used the same proportion for both batches of oatmeal.



What's More

- 1) Carol finds a number of cars in a parking lot. Complete the following table.



Number of cars	1	2	3	5	15
Number of wheels					

The ratio of the number of cars to the number of wheels is ____ : ____.

If there are 12 wheels, then there are ____ cars.

If there are 36 wheels, then there are ____ cars.

The number of ____ and the number of ____ are in the same proportion.

- 2) Tony counts the number of square-shaped and heart-shaped biscuits in a container. Complete the following table.



Number of squares	2	4		8	
Number of hearts		10	15		25

The ratio of the number of squares to the number of hearts is ____ : ____.

The _____ and _____ are in the same _____.

For every 10 square-shaped biscuits, there are ____ heart-shaped biscuits.

- 3) Members of a volunteer organization are divided into groups for different projects. The table shows the number of males and females in each group. Complete the table.

Number of males		9	12		18
Number of females	8		16	20	

The ratio of the number of males to the number of females is ____ : ____.

If there are 32 females in a group, then there are ____ males.

What can you say about the number of males and the number of females?



What I Have Learned

- ❖ **Ratio** is a comparison of two or more quantities.
- ❖ Two equivalent ratios form a **proportion**.

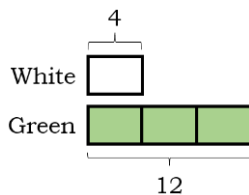


What I Can Do

- 1) Draw a block model to illustrate each situation. Then, write the proportion.

Example:

The number of white and green marbles in a bag are in the ratio 1 : 3.
There are 4 white marbles and 12 green marbles in the bag.



$$\frac{\text{number of white marbles}}{\text{number of green marbles}} = \frac{1}{3} = \frac{4}{12}$$

- A. The ratio of the number of 5-peso coins to the number of 10-peso coins in a bottle is 7 : 4. There are 49 5-peso coins and 28 10-peso coins.
- B. Ben and Sheila has some money in the ratio 6 : 5. Ben has ₱330 and Sheila has ₱275.
- C. A painter mixed 50 ml of yellow paint with every 90 ml of blue paint to obtain green paint. She used 450 ml of blue paint and 250 ml of yellow paint.
- 2) Ricky used cement and sand in the ratio 5 : 3 to make different mixtures.

- A. Draw a block model to show the ratio.
- B. Use the same proportion to fill in the table.

Mixture	A	B	C	D
Number of pails of cement	5			
Number of pails of sand	3			

- C. Use the information in the table to write five different proportions.
- D. If Ricky used 27 pails of sand, how many pails of cement did he use?



Assessment

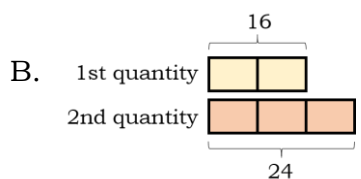
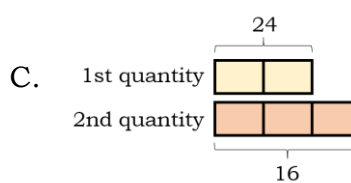
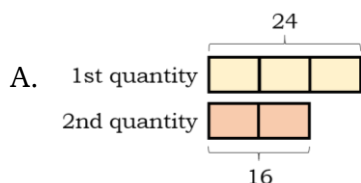
Read and analyze each item carefully. Then choose the letter of the best answer.

- 1) Which statement represents the following model?

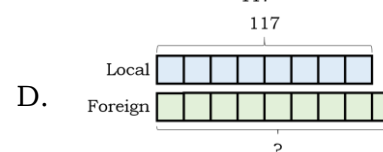
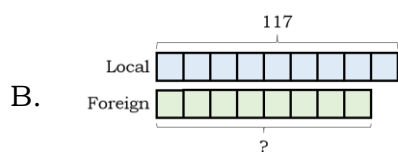
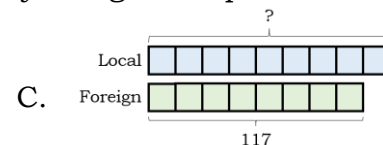
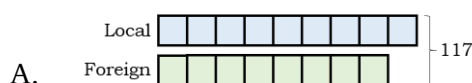


- A. $12 : 6 = 2 : 1$
 B. $2 : 4 = 12 : 6$
 C. $4 : 8 = 6 : 12$
 D. $12 : 6 = 4 : 2$

- 2) Which block model shows $3 : 2 = 24 : 16$?



- 3) Thalia has 117 local stamps. She has 9 local stamps for every 8 foreign stamps. Which model can help find how many foreign stamps she has?



- 4) In two games, the number of points scored by a basketball player for every number of shots he took were in the same proportion. Which of the following does NOT show the correct proportion?

	Game 1	Game 2
Number of points	12	18
Number of shots	14	21

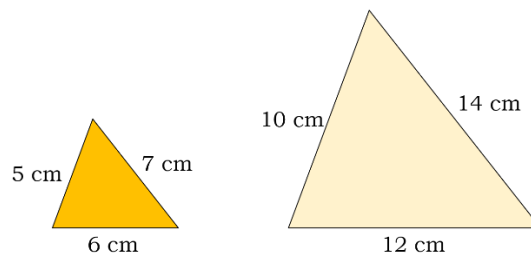
- A. $12 : 14 = 6 : 7$
 B. $12 : 21 = 18 : 14$
 C. $6 : 7 = 18 : 21$
 D. $12 : 14 = 18 : 21$

- 5) Which of the following statements is TRUE?
- A. In a proportion, the quantities can be written in any order.
 - B. When the ratios of two quantities are different, the two quantities are in the same proportion.
 - C. A proportion is a statement that two ratios are equivalent.
 - D. When the ratios of two quantities are the same, the two quantities are not in proportion.



Additional Activities

- 1) Imagine you have some blue and yellow buttons.
- A. Form 5 different groups of buttons. Make sure that the ratio of the number of blue buttons to the number of yellow buttons in each group is 4 : 3.
 - B. Construct a table to record your results in (a).
 - C. Use the information in the table to write five (5) proportions.
 - D. If there are 72 yellow buttons in a group, how many blue buttons are there?
- 2) On a bulletin board, a teacher posted the following triangles.



A pupil said that the sides of the triangles are not in proportion.
Do you agree? Why or why not?

- 3) Ratio A is equivalent to Ratio B. Ratio B is equivalent to Ratio C.
Do Ratios A and C form a proportion?
Explain your thinking.



Answer Key

What I Know

A.

Amount of salt	3	2	4	6	9	12	15
Amount of sugar	2	3	4	6	9	12	15

1) Amount of salt: 15, Amount of sugar: 10

2) Amount of salt: 10, Amount of sugar: 15

3) Possible answers:

1) F, 2) T, 3) F, 4) T, 5) T

B.

1) F, 2) T, 3) F, 4) T, 5) T

3) Possible answers:

1) F, 2) T, 3) F, 4) T, 5) T

6 : 4 = 9 : 6, 6 : 4 = 12 : 8, 6 : 4 = 15 : 10

9 : 6 = 12 : 8, 9 : 6 = 15 : 10, 12 : 8 = 15 : 10

What's In

Possible answers:

1) The ratio of the number of seashells Leah collected to the number of seashells Joy collected is 8 : 7.
 2) The ratio of the number of seashells Joy collected to the total number of seashells both girls collected is 7 : 15.
 3) The ratio of the total number of seashells both girls collected to the number of seashells Joy collected is 15 : 7.
 4) The ratio of the number of seashells Leah collected to the total number of seashells both girls collected is 8 : 15.
 5) The ratio of the total number of seashells both girls collected to the number of seashells Leah collected is 15 : 8.
 6) The number of seashells Joy collected is $\frac{8}{7}$ of the number of seashells Leah collected.
 7) The number of seashells Leah collected is $\frac{8}{7}$ of the number of seashells Joy collected.
 8) The number of seashells Joy collected is $\frac{15}{7}$ of the total number of seashells both girls collected.
 9) The number of seashells both girls collected is $\frac{7}{15}$ of the number of seashells Joy collected.
 10) The number of seashells Leah collected is $\frac{15}{8}$ of the total number of seashells both girls collected.
 11) The number of seashells both girls collected is $\frac{8}{15}$ of the number of seashells Leah collected.

(I



$$\frac{\text{number of 5-peso coins}}{\text{number of 10-peso coins}} = \frac{7}{4} = \frac{49}{28}$$



$$\frac{\text{Ben's money}}{\text{Sheila's money}} = \frac{5}{6} = \frac{\text{£275}}{\text{£330}}$$



$$\frac{\text{amount of yellow paint}}{\text{amount of blue paint}} = \frac{5}{9} = \frac{250}{450}$$

(I

Number of cars	1	2	3	5	15
Number of wheels	4	8	12	20	60

The ratio of the number of cars to the number of wheels is **1 : 4**.

If there are 12 wheels, then there are **3** cars.

If there are 36 wheels, then there are cars. **6**

The number of **cars** and the number of **wheels** are in the same proportion.

Number of squares	2	4	6	8	10
Number of hearts	5	10	15	20	25

The ratio of the number of squares to the number of hearts is **2 : 5**.

The number of squares and the number of hearts are in the same proportion.

For every 10 square-shaped biscuits, there are **25** heart-shaped biscuits.

Number of females	8		9	12		18
Number of males						

The ratio of the number of males to the number of females is **3 : 4**.

If there are 32 females in a group, then there are **24** males.

What can you say about the number of males and the number of females?

The number of males and the number of females in each group are in the same proportion.

What I Can Do

2)

Cement 

Sand 

a. Answers may vary as long as the ratio for each mixture is equivalent to 5 : 3.)
 b. (Answers depend on the given values in the table in (b).)
 c. 45
 d. 45

Assessment

1) D
 2) A
 3) B
 4) B
 5) C

Additional Activities

1) a-b (Answers may vary as long as the ratio of the number of blue and yellow buttons in each group is 4 : 3, and that five (5) groups of buttons are recorded in the table.)
 c) (Answers depend on the given values in the table in (b).)
 d) 96

2) No. (Explanation must show understanding that the ratios of the pairs of corresponding sides are the same.)

3) Yes. (Explanation must show understanding of proportion by comparing Ratios A, B and C.)

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